



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



IntellProcess Group
Where Process Meets Intelligence.

Process Engineering Specialized Courses

Instructors:

- ❖ **Dr. Ali Mohammad Sahlodin**
Faculty Member, Amirkabir University of Technology (Tehran Polytechnic) (Former Senior R&D Engineer at Aspen Technology Inc)
- ❖ **Sajjad Koohi**
Senior Process Engineer
- ❖ **Mohammad Hadi Alizadeh**
Senior Process Engineer

Contact: sahlodin@aut.ac.ir | +982164543154

Prog

1. Steady-state Simulation with ASPEN HYSYS
2. Dynamic Simulation with ASPEN HYSYS
3. Relief Load Calculation and Overpressure Protection
4. Tank, Vessel, and Tower Design
5. Flare Network Design with FlareNet & FlareSim
6. Process Diagram Development – PFD, P&ID
7. Programming for Process Engineering
8. Adsorption Column Design with Aspen Adsim
9. Air Cooler Design with HTRI
10. Shell & Tube Design with HTRI
11. Steady-state Simulation with ASPEN PLUS
12. AI for Process Engineering



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Steady-State simulation with ASPEN HYSYS

Description:

Comprehensive training on steady-state process simulation using ASPEN HYSYS, covering flow sheet development, unit operations (distillation, heat exchangers, reactors, compressors, pumps), fluid package selection, oil characterization, hypo component generation, crude oil blending, recycle convergence, column hydraulic rating, reactor kinetics, sensitivity analysis, optimization, spreadsheet, and validation.

Software:
ASPEN HYSYS

Course Duration:
24 Hours

02

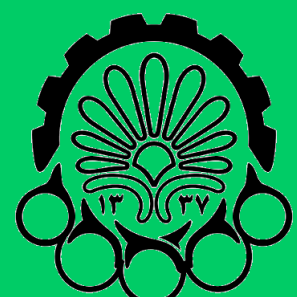
Dynamic simulation with ASPEN HYSYS

Description:

Comprehensive training on dynamic simulation using ASPEN HYSYS, covering dynamic model setup from steady-state basis, unit operations (distillation, heat exchangers, reactors, compressors, pumps), pressure-flow network configuration, control system design and tuning (PID controllers), safety shutdown and interlock systems, disturbance analysis and scenario testing, dynamic column and reactor behavior, compressor surge analysis and anti-surge control, emergency depressurization studies, controller performance validation, and validation against plant operational data.

ASPEN HYSYS

DYNAMIC SIMULATION



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Software:
ASPEN HYSYS

Course Duration:
24 Hours

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Relief Load Calculation and Overpressure Protection

Description:

Comprehensive training on relief load calculation and overpressure protection, covering relief system design principles, identification of overpressure scenarios (blocked outlet, fire exposure, thermal expansion, cooling failure, power failure, instrument failure, reflux failure), API 520/521/526 standards, pressure relief device selection (PSV, PRV, rupture discs), PSV sizing and orifice calculation, rupture disc selection and application, inlet and outlet piping configuration and pressure drop analysis, and depressurization and blowdown system design.

Software:
ASPEN HYSYS-Safety Analysis Module

Course Duration:
24 Hours



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Tank, Vessel, and Tower Design

Description:

Comprehensive training on process design of storage tanks, covering atmospheric and pressurized storage tanks including fixed roof, floating roof, and spherical tanks, with topics including tank selection, tank sizing, overpressure protection, and blanketing, process design of pressure vessels, covering vertical and horizontal vessels including separators (two-phase and three-phase), with topics including vessel sizing, process internals selections, overpressure protection, and process design of process towers, covering trayed and packed towers, with topics including tower sizing, hydraulic design, tray and packing selection, process internals, stage efficiency, flooding and entrainment analysis, and pressure drop calculations.

Software:
KOCH-GLITCH/EXCEL

Course Duration:
30 Hours

05

Flare Network Design with FlareNet & FlareSim

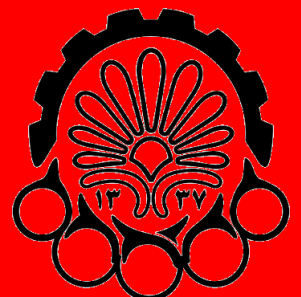
Description:

Comprehensive training on flare network and stack design, covering tail pipe sizing, sub-header and header sizing, back pressure calculations, flare location determination, knock-out drum sizing, flare pump sizing, radiation analysis, and API 521 & API 537 compliance.



Software:
ASPEN Flarenet/FlareSim

Course Duration:
18 Hours



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Process Diagram Development – PFD, P&ID

Description:

Comprehensive training on the development and interpretation of PFD, P&ID, and UFD, covering project specifications, Legend & Symbols, IPS-E-PR-170 and IPS-E-PR-230, PFD components, Heat & Material Balance, equipment data requirements (Heat Exchanger, Furnace, Reactor, Column, Pump, Compressor, Tank), instrumentation and control systems, line information and PMS, HYSYS line sizing, Shutdown levels, UFD development, and case studies for gas and oil refinery units.

Software:
AUTOCAD

Course Duration:
24 Hours

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Programming for Process Engineering

Description:

Comprehensive training covering Python programming for data analysis and visualization, MATLAB for numerical computing and algorithm development, and gPROMS for steady-state and dynamic process modeling. Includes model development for unit operations (Reactors, Distillation Columns, Heat Exchangers, Pumps, Compressors, Crystallizers, Bioreactors), parameter estimation, process optimization, heat and material balance, equipment data requirements, instrumentation and control systems, and dynamic optimization for batch and continuous processes. Features case studies for oil and gas, petrochemical, and pharmaceutical bioprocesses, along with integration workflows combining Python, MATLAB, and gPROMS for end-to-end simulation and analysis.

Software:
Python/MATLAB/gPROMS

Course Duration:
30 Hours

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08

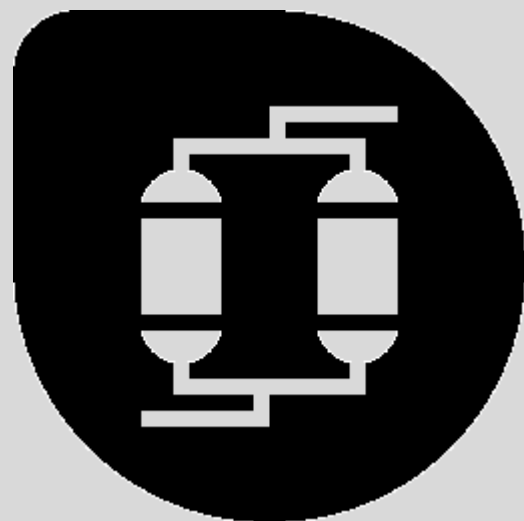
Adsorption Column design with Aspen Adsim

Description:

Comprehensive training on simulation of PSA, VSA, and fixed-bed adsorption using Aspen Adsim, covering component/adsorbent definition, equilibrium isotherms, mass transfer coefficients (LDF), mass and energy balances, cycle simulation, configuration, bed interactions, Cyclic Steady State determination, breakthrough curve analysis, and industrial case studies, including air separation unit.

Software:
ASPEN Adsim

Course Duration:
12 Hours



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HTRI



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Air Cooler Design with HTRI

Description:

Comprehensive training on air cooler design and simulation, covering thermal design, hydraulic design, fin tube selection, fan selection (Force Draft / Induced Draft), tube bundle configuration, header design, air-side and process-side modeling, pressure drop calculations, vibration analysis, performance rating, troubleshooting, and optimization.

Software:
HTRI

Course Duration:
24 Hours

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HTRI



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Shell & Tube Design with HTRI

Description:

Comprehensive training on steady-state process simulation using ASPEN HYSYS, covering flow sheet development, unit operations (distillation, heat exchangers, reactors, compressors, pumps), fluid package selection, oil characterization, hypo component generation, crude oil blending, recycle convergence, column hydraulic rating, reactor kinetics, sensitivity analysis, optimization, troubleshooting, and validation.

Software: HTRI

Course Duration: 24 Hours

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Steady-state simulation with ASPEN PLUS

Description:

Comprehensive training on steady-state process simulation using ASPEN PLUS, covering flowsheet development, unit operations (distillation, heat exchangers, reactors, columns, flash drums, pumps, compressors, mixers, splitters), property method selection, petroleum characterization and pseudo-component generation, phase envelope curve generation, physical property adjustment and regression, electrolyte chemistry, solids handling, polymer processing, rate-based distillation, rigorous column modeling (RadFrac), column hydraulic rating (Tray Sizing, Packed Sizing), batch distillation, reactor kinetics, sensitivity analysis, optimization, design specification, case studies, calculator blocks, equation-oriented (EO) modeling, convergence troubleshooting, and validation against plant data.

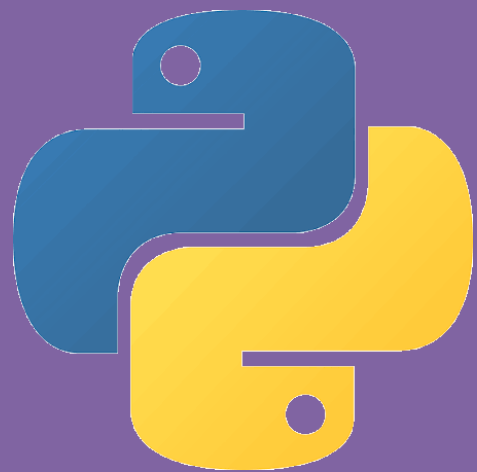
Software:
ASPEN PLUS

Course Duration:
24 Hours

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AI for Process Engineering

Description:

Comprehensive training on AI and machine learning in process engineering and automation, covering fault detection and diagnosis, soft sensing and virtual sensor development, data-driven modeling, reinforcement learning for process control and optimization, predictive maintenance, anomaly detection, time-series forecasting, hybrid modeling (combining machine learning with process simulation), automation engineering for control system integration, and leveraging Large Language Models (LLMs) for report generation and engineering assistance.

Software:
Python

Course Duration:
18 Hours

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THANK YOU



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